MIL-M-40024B (DSA-DM)
31 October 1975
SUPERSEDING
MIL-M-0040024A
20 September 1965

MILITARY SPECIFICATION

MATTRESS, STRETCHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of latex-foam rubber mattress with case, suitable for use with an ambulance stretcher.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-276	Thread, Cotton
TT-I-1795	Ink, Marking, Stencil, Opaque, (Porous and Non-Porous Surfaces)
ZZ-M-91	Mattress, Bed, Latex Foam
CCC-C-700	Cloth, Coated, Vinyl Coated (Artificial Leather)
DDD-L-20	Label, for Clothing, Equipage and Tentage (General Use)

Military

MIL-M-3996	Mattresses - except Pneumatic Types, Packaging and Packing of
MIL-R-5001	Rubber Cellular Sheet, Molded and Hand Built Shapes, Latex Foam

FSC 6530

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STANDARDS

Federal

FED-STD-751 Stitches, Seams, and Stitchings

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1564 Slab Flexible Urethane Foam

(Application for copies should be addressed to the American Society for Testing and Materials 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

* 3.1 Materials. All materials shall be new, previously unused.

3.1.1 Latex-foam. Pin core latex-foam shall conform to the material requirements for latex-foam in ZZ-M-91. Cored latex-foam shall conform to type I, class medium, grade A, and the uncored latex-foam border to type II, class medium, grade A of MIL-R-5001.

* 3.1.2 Polyether foam. Polyether foam, when used as the filler, shall have a density of at least 3 pounds per cubic foot and an indentation load deflection (ILD) from 31 to 36 pounds of force per 50 square inches area to produce a 25% indentation, when tested as specified in 4.4.1.

* 3.1.3 Artificial leather. The artificial leather for the mattress case shall conform to Class 2, and shall be fire-resistant treated in accordance with Treatment a (1), of CCC-C-700. The color of the artificial leather shall be brown. The artificial leather shall be finished with a cowhide or other similar fine-grain finish. As an alternate, material for the mattress case shall be 6 ounces per square yard, minimum, green, fire-resistant, vinyl-coated woven nylon fabric, with appropriate thread color.

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* 3.1.4 Thread. The cotton thread for seaming and stitching shall be four cord, ticket No. 12, conforming to type IB3 (for needle thread) and type IA3 (For bobbin thread) of V-T-276. The thread shall be dyed to match russet K (Cable No. 66025) of TCA Cable Numbers and Colors. As an alternate, polyester thread shall be used for stitching.

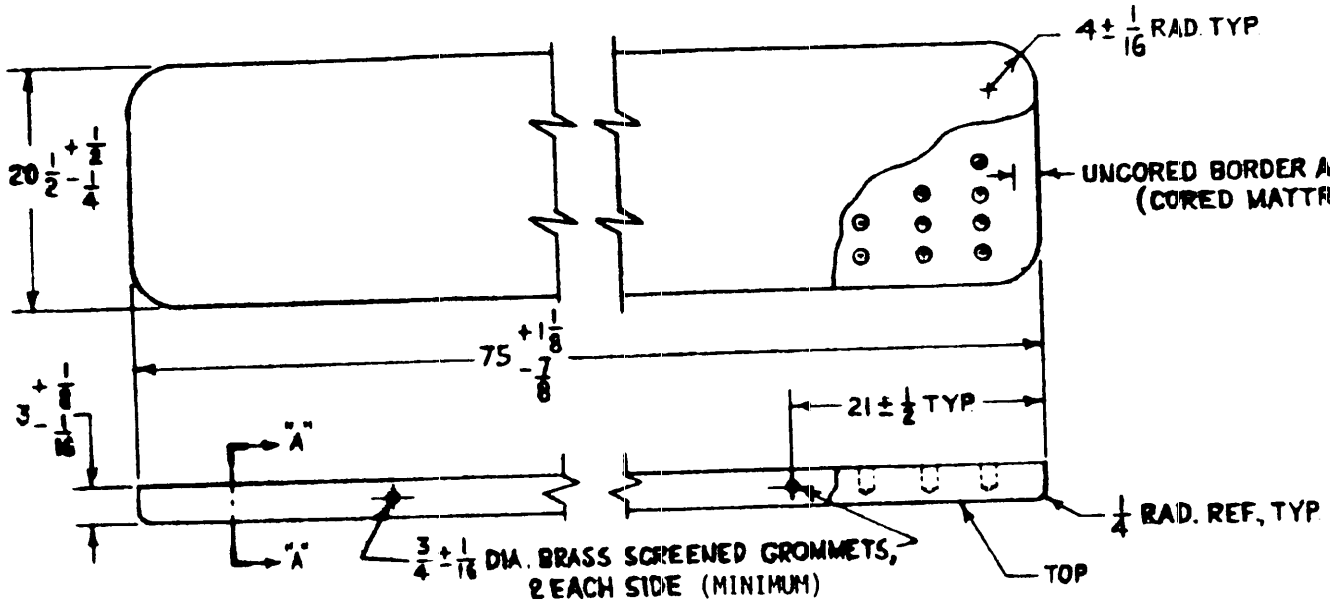
* 3.2 Design and construction. The mattress and case shall conform to Figure 1. The latex-foam mattress shall be of one-piece finished construction. It shall be made of one solid piece of pin-core or cored latex, or it may be molded in not more than four sections, exclusive of border, and bonded together in such a manner that the seams shall not be so firm as to be detectable through the case. The corners shall be rounded. If cored latex is used, the core holes shall not extend within 3/8 inch of the outer face and there shall be an uncored latex-foam edge border 1-1/16 inches wide, premolded or cemented on all sides of the cored latex-foam mattress.

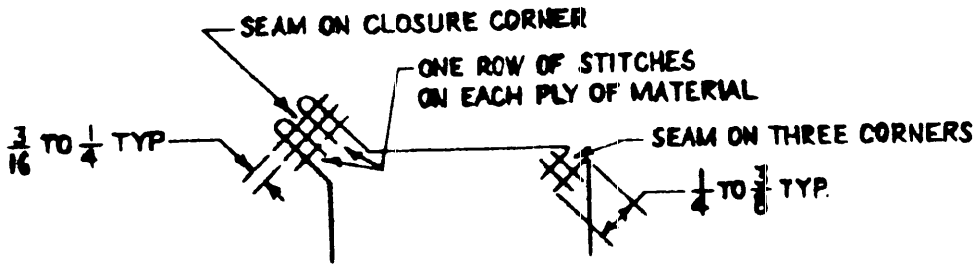
3.2.1 Mattress case. Each finished mattress case shall fit snugly over the latex-foam mattress, but shall not pucker or pull the mattress out of line due to an excessively close fit. The top and bottom panels of the mattress case shall each be cut in one piece. They shall be joined to the box edge panel at the seams. The box shall be formed of one continuous piece to be joined at the center of one end of the mattress. As an alternate, the box shall be formed of four pieces, and a slide fastener shall be inserted half way between the upper and lower seams. Each side of the box panels shall be provided with not less than two round brass screened ventilators.

* 3.2.1.1 Stitches, seams and stitching. All stitches, seams, and stitchings shall conform to FED-STD-751. Stitch type 301 with 6 to 9 stitches to the inch shall be used throughout. Seam type shall be SSa-2, and the closing end shall be seamed with seam type SSc-2. As an alternate, the top and bottom panels shall be entirely seamed to the box edge with seam type SSa-2 and closure shall be made by a slide fastener.

* 3.2.1.2 Label. Each mattress shall have an identification label, made of cloth, coated and printed in accordance with Type I of DDD-L-20, and securely attached. It shall be caught into the box edge seams at the center of one end of the box edge panel. As an alternate, the label may be positioned on a side. The label shall include the contract or purchase order number.

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SECTION "A-A"
(COVER ONLY)

NOTE: ALL DIMENSIONS ARE IN INCHES

MATTRESS AND COVER

FIGURE 1

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3.2.1.3 State legal requirement. Additional markings as required by state health regulations, shall be affixed, as directed by laws of the state in which the mattresses were manufactured.

* 3.2.2 Marking. Each latex-foam mattress shall be permanently marked with the manufacturer's name or registered trademark, and year of manufacture, so that the source and time of manufacture may be readily determined. Black printing ink or black stencil ink which meets the light and water spray fastness requirements as specified in TT-I-1795, shall be used for all marking.

* 3.3 Workmanship. The workmanship shall be first class in every respect and free from imperfections in manufacture which might affect either the appearance or serviceability. Molded or assembled shapes which, due to manufacturing conditions may have to be altered physically, will be accepted under this specification provided such alteration does not affect the serviceability, the size, or the shape beyond tolerances specified herein. The material used in such repair or alteration shall be of composition and quality identical with the original product. Mattress shall be free of objectionable odor.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and the following.

4.2.1.1 Visual and dimensional examination of latex-foam unit. The unit of product for sampling shall be a completely fabricated latex foam unit prior to insertion in the case. Sampling shall be in accordance with Table I.

Table I. Sampling for examination of latex-foam unit

	Inspection level	AQL (percent defective)
For visual examination	S-4	4.0
For dimensional examination	S-4	4.0

4.2.1.2 Visual and dimensional examination of end item. The unit of product for sampling shall be one completely-fabricated ambulance stretcher mattress. Sampling shall be in accordance with Table II.

* Table II. Sampling for examination of completed mattress

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	I	2.5
Total (major and minor combined) defects	I	6.5
For dimensional examination	S-3	4.0

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* 4.2.2 For test. Sampling for the test specified in 4.4.1, to satisfy the requirement of 3.1.2, shall be conducted in accordance with MIL-STD-105. The unit of product for sampling shall be a completely fabricated, latex-foam unit, or a cross-section at least 15 inches square. The inspection level shall be one unit per lot, and the acceptance number shall be zero.

4.3 Examination. The mattress shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects.

* 4.3.1.1 Latex-foam unit. The latex-foam unit shall be examined for defects in material, construction, workmanship, and marking in accordance with Table III. Examination shall not be restricted to the classified possible defects listed below.

Table III. Classification of defects, latex-foam unit

Category	Defects
Major	
101	Any malformation which may affect serviceability.
102	Any tear not cemented together.
103	Material has large void.
104	Material not new. (Has previously been used.)
105	Finished unit not in one piece.
106	Any alteration which may affect serviceability.
107	Material used in alteration not same as basic latex-foam.
108	Material has objectionable odor.
109	Edge border missing or not uncored latex-foam.
110	Edge border not attached to cored portion as specified.
111	Edge border not on all four sides of cored unit structure.
112	Identification marking missing, incorrect, illegible, incomplete, or not black in color.

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* 4.3.1.2 End item. The completely fabricated ambulance stretcher mattress shall be examined for defects in material, construction, workmanship and marking in accordance with Table IV. Examination shall not be restricted to the classified possible defects listed below.

Table IV. Classification of defects, completely fabricated mattress

Category	Defects
Major	
101	Fabric not type specified.
102	Fabric torn, cut, or contains holes.
103	Color not as specified.
104	Coating peeling, blistered, split or cracked to an extent affecting serviceability.
105	Stitching gage less than 1/4 inch or more than 5/16 inch.
106	Stitch type other than 301.
107	Any open seam on both rows of stitching.
108	Any seam on top or bottom panel of mattress covering.
109	Any ventilator missing (four required, two on each side).
110	Ventilator does not have screened opening.
111	Material has objectionable odor.
Minor	
201	Fabric patched or mended.
202	Fabric has shade variations affecting appearance.
203	Not matching shade between panels or boxing.
204	Coating stained, streaked or spotted affecting appearance.
205	Coating peeling, blistered, split or cracked.
206	Coating damaged, scratched or scuffed affecting appearance.
207	Fabric creased or wrinkled affecting appearance.
208	Fabric dirty or greasy to an extent that cannot be completely and easily removed with a damp cloth.
209	Edges of mattress out of alignment to an extent affecting appearance.
210	Not seamed with seam type SSa-2 (except closing end).
211	Closing end not seamed with seam type SSc-2.
212	Seam open on one row of stitching.
213	Seam pleated badly or twisted.

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Table V. Classification of defects, completely fabricated mattress (Cont'd)

Category	Defects
Minor	
214	Seam uneven or not neatly accomplished, affecting appearance.
215	Seam in wrong location.
216	Five stitches per inch or less (check both seam types carefully).
217	Ten stitches per inch or more (check both seam types carefully).
218	Needle chews.
219	Broken stitch not resulting in an open seam.
220	More than two skipped stitches or runoffs.
221	Raw edge of fabric noticeable at seam.
222	More than one thread end not trimmed off.
223	Color of sewing thread not similar to basic color of covering fabric.
224	Boxing not formed of one or four pieces of material, as specified.
225	Ends of boxing not joined at the center of one end of the mattress (if single piece boxing).
226	Covering too loose or puckered to an extent affecting appearance.
227	Furrow, lump, or bulge affecting appearance.
228	Cemented seam of foam rubber can be felt through the artificial leather.
229	Ventilator not of type or not fabricated of material specified.
230	Ventilators incorrectly spaced or improperly located.
231	Label(s) missing.
232	Label(s) not attached as specified or not in location specified.
233	Label(s) insecurely attached.
234	Text of label(s) incomplete, incorrect or illegible.

4.3.1.3 Examination for dimensions. The latex foam unit, the cover, and the complete stretcher mattress shall be examined for dimensions. Any dimension that does not fall within the specified tolerances shall be classified as a defect.

4.4 Tests.

4.4.1 Indentation load deflection. The sample (4.2.2) shall be tested for indentation load deflection (ILD) as specified in ASTM D1564.

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* 5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing, and marking for the levels specified shall be in accordance with MIL-M-3996. One (1) mattress shall be packed per exterior container.

6. NOTES

6.1 Intended use. The mattress is intended for use with 6530-00-982-2042, Stretcher, Ambulance.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1).

6.3 This specification covers the following item appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-784-2120	MATTRESS, STRETCHER, 74 by 20 by 3 inches

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DSA-DM

Project No. 6530-1360

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DOCUMENT IDENTIFIER AND TITLE

MIL-M-40024B (DSA-DM) MATTRESS, STRETCHER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

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